

# Virtual Fashion Fit: An AI-Driven 2D Virtual Try-On System for Traditional Pakistani Clothing

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## Abstract

The Virtual Fashion Fit project addresses a critical gap in virtual try-on systems for traditional Pakistani clothing, such as kurtas and shalwar kameez. Existing online visualization solutions often lead to customer distrust due to the absence of tailored solutions, resulting in distorted renderings with issues like squeezed sleeve and waist textures. This project develops an AI-driven virtual try-on system leveraging the Sequential Deformation Virtual Try-On (SD-VITON) model (Liu et al., 2023) to deliver realistic 2D visualizations, thereby enhancing customer confidence. The system is deployed on Hugging Face Spaces, ensuring accessibility across low-hardware devices, and utilizes Python (3.9), PyTorch (2.0), OpenCV (4.8), and CUDA (11.8) for efficient processing. The proposed system demonstrates effective virtual try-on results for traditional garments as illustrated in Figure 1.

**Keywords:** Virtual Try-On, Traditional Clothing, Shalwar Qameez, AI, Computer Vision, SD-VITON, 2D Visualization

## 1 Introduction

The rapid growth of e-commerce has transformed the fashion retail landscape, yet a significant challenge persists: the inability for customers to physically try on garments before purchase. This limitation is particularly pronounced for traditional clothing, like the shalwar qameez, where fit, drape, and cultural nuances are paramount. Generic virtual try-on solutions often fall short for such attire, leading to unrealistic renderings and customer dissatisfaction. This project, "Virtual Fashion Fit," aims to bridge this gap by providing a specialized virtual try-on system designed specifically for traditional Pakistani clothing.

Our system is an updated form of the SD-VITON model (Liu et al., 2023), adapted and optimized to address the unique requirements of traditional wear. By providing realistic 2D visualizations, Virtual Fashion Fit seeks to enhance customer confidence in online purchases, reduce return rates, and offer an unparalleled level of convenience. This paper details the methodologies employed, including data preprocessing, the core model architecture, challenges encountered, and future enhancements, demonstrating the feasibility and effectiveness of an AI-driven approach to virtual try-on for cultural garments.

## 2 Related Work

Existing virtual try-on research has explored various techniques, ranging from 2D image manipulation to 3D garment reconstruction. Models like SD-VITON (Liu et al., 2023) have shown promising results in general virtual try-on scenarios. However, these models are typically trained on datasets like Zalando, which predominantly feature Western clothing styles (Figure 2). The complexities of traditional garments, such as intricate embroidery, specific drapes (e.g., dupatta), and varying fabric textures, present unique challenges that generic models may not adequately address, often leading to distorted renderings.

This project specifically focuses on adapting and improving upon the SD-VITON framework to handle the nuances of traditional Pakistani attire, recognizing the need for tailored solutions to achieve realistic and trustworthy virtual try-on experiences for this specific clothing category.

## 3 Methodology

The Virtual Fashion Fit system builds upon the Sequential Deformation Virtual Try-On (SD-VITON) model (Liu et al., 2023). The overall architecture in-



Figure 1: Results of our Virtual Fashion Fit Model.

volves several key preprocessing steps to prepare both the person image and the garment image for the virtual try-on process.

### 3.1 Dataset and Preprocessing

Our dataset was gathered from commercial sites, which typically provide images of people wearing clothes rather than separate garment images. This necessitated a robust preprocessing pipeline to extract the clothing from the person images.

- **Cloth Extraction:** We evaluated several models for extracting clothing from the person images: Hugging Face U2 Net (Hugging Face. (n.d.)), Mask R-CNN (He et al., 2017), and Segment Anything Model (SAM) (Kirillov et al., 2023). Comparative results, as shown in Figure 2 (d), indicated that SAM offered superior performance in segmenting both the shalwar and qameez compared to Hugging Face U2 Net and Mask R-CNN. However, SAM generates multiple segments, requiring manual selection of the appropriate cloth segment from all generated segments. This process creates separate cloth masks for the try-on process.
- **Human Keypoints:** To understand the person’s pose, we utilized MoveNet to generate COCO-format human keypoints, similar to how OpenPose generates them, as shown in Figure 3a. These keypoints are crucial for guiding the deformation of the virtual garment.
- **Densepose:** The original Densepose model used by SD-VITON (Liu et al., 2023) was employed for generating densepose maps, as shown in Figure 21, providing expected results that detail surface correspondences between the human body and a 3D model.

- **Human Parsing:** This is a critical step for the main model as it meticulously segments the input image into distinct human body parts, creating an “agnostic” representation of the person’s shape and pose as shown in Figure 3c. This detailed anatomical map is essential for accurately guiding the deformation of the virtual garment to fit the user’s body, enabling realistic warping and seamless blending of the new clothing, ensuring correct layering and occlusion for a believable try-on result. Without accurate human parsing, the garment could be warped incorrectly, leading to poor fit, unrealistic stretching, or parts of the garment appearing in the wrong place. While SD-VITON (Liu et al., 2023) originally used CIHP (Luo et al., 2020) for human parsing, its processing time of 10-20 minutes per image was impractical for a real-time experience. To make the real-time experience more exciting, we needed to minimize the time to seconds. We therefore explored alternative models for human parsing and selected SCHP (Li et al., 2019) as illustrated in Figure 2c, which usually takes a few minutes to parse the whole dataset.
- **Agnostic Image:** The primary goal of agnostic-v3.2 is to extract the underlying body shape and pose of the person in the image without being influenced by the texture, color, or style of the clothes they are currently wearing. The model uses this information to understand how the new clothing item (which the user wants to “try on”) should be deformed, bent, and stretched to fit the person realistically. SD-VITON’s (Liu et al., 2023) sequential deformation process heavily relies on this agnostic input to predict the appearance flow correctly. By using an agnostic representation of the person, the model avoids “texture leakage” or interference from the original clothes. For instance, if the person is wearing a patterned shirt in the input photo,

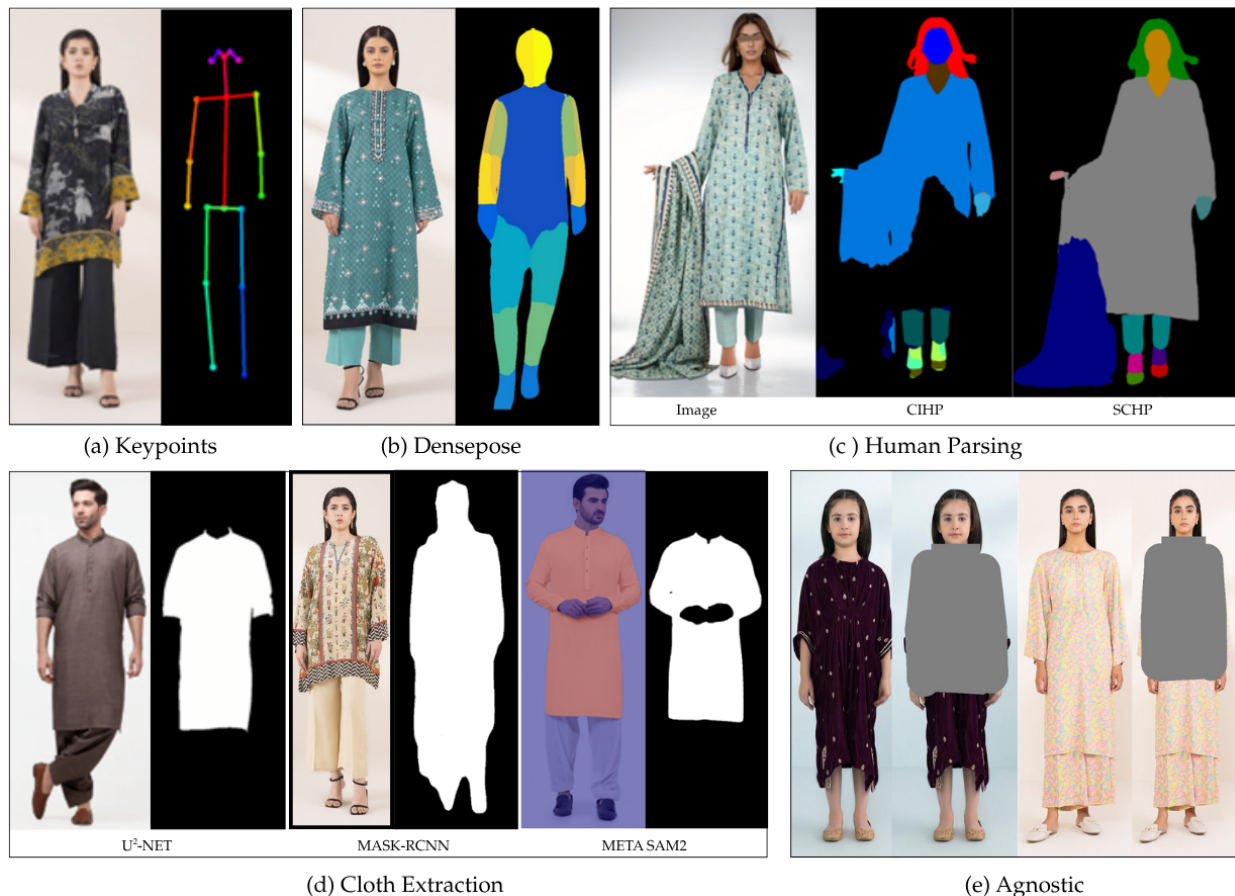


Figure 2: Visualizations of preprocessing steps: (a) MoveNet for generating COCO-format keypoints for human image. (b) Densepose generated using the same model as SD-VTON. (c) Original Image and a comparison between CIHP (Luo et al., 2020) and SCHP (Li et al., 2019) for Human Parsing. (d) Results of models for cloth extraction (U2net, Mask R-CNN (He et al., 2017), and Meta’s SAM (Kirillov et al., 2023)). (e) Agnostic-v3.2 shows the agnostic mask for the model used.

the model shouldn’t let that pattern show through or influence the try-on of a new, different garment. The agnostic mask for the model used is shown in Figure 2e.

### 3.2 Model Implementation

The core try-on functionality is based on the SD-VITON (Liu et al., 2023) model. The system is implemented using Python (3.9) with PyTorch (2.0) for deep learning operations, OpenCV (4.8) for image processing, and CUDA (11.8) for GPU acceleration. The system is deployed on Hugging Face Spaces, ensuring broad accessibility even for users with low-hardware devices.

## 4 Results

The Virtual Fashion Fit model demonstrates promising results in virtually trying on traditional Pakistani clothing (Figure 3). The system effectively overlays selected garments onto a person’s image, providing a visualization of the fit.

Figure 1 illustrates several successful try-on instances, showcasing the model’s ability to integrate different shalwar qameez designs onto various individuals. The visual quality of the try-on images is notably improved compared to generic solutions, particularly in terms of how the fabric appears to conform to the body shape.

Table 1: Ablation Study of Virtual Fashion Fit System Components. (Lower values for FID, KID, LPIPS are better; Higher values for SSIM are better.)

| Configuration                      | FID↓     | KID↓   | LPIPS↓ | SSIM↑  |
|------------------------------------|----------|--------|--------|--------|
| A. Full Virtual Fashion Fit System | 106.2830 | 0.0094 | 0.4787 | 0.7758 |

## 5 Challenges and Limitations

Despite the promising results, the project encountered several challenges and limitations:

- **Resultant Cloth Length:** A significant glitch observed is that the resultant image cloth will have the length of the original image cloth, not the cloth that is being tried out. This issue stems from the characteristics of the generated agnostic mask during preprocessing, which dictates the overall body shape and proportions.
- **Garments with Extra Cloth:** Some cloth may contain extra cloth (e.g. Dupatta in hands or in neck, strawler on head), these can cause results to distort, as illustrated in Figure 4. For example, in Figure 4a, the selected cloth contains a dupatta in hand, which can cause serious distortions in the results, as the cloth is not properly deformed. Similarly, if the selected person image has a dupatta in hand, there is a slight distortion in the results as the hand is deformed, and also there is a disruption in the shalwar. The results may be affected by images with complex patterns or any extra cloth.
- **Agnostic Mask Generation Failures:** For some images, the agnostic-v3.2 could not be created for unknown reasons. This directly impacts the ability to process those images for virtual try-on, as the agnostic mask is crucial for guiding garment deformation.
- **No Separate Cloth Images:** One thing is that we don't have the image of the cloth separately. We've run models on extracting the cloth from the person's image and creating cloth-masks of them. While this approach is effective, it adds a layer of complexity compared to having pre-segmented garment images.
- **Orientation Mismatch:** If the input person image and the selected cloth image have significantly different orientations (e.g., person facing slightly left, garment facing perfectly front), achieving a seamless and realistic try-on becomes difficult. The 2D deformation models may struggle to correctly align and deform the garment, leading to misaligned features or unnatural twisting.

## 6 Future Enhancements

Building upon the current implementation, several avenues for future enhancement can be explored:

- **Improved Agnostic Mask Generation:** Addressing the issue of the resultant cloth length and the inability to generate agnostic masks for some images is paramount. This could involve exploring more robust human parsing and agnostic mask generation models, potentially incorporating techniques that allow for more flexible length adjustments based on the selected garment.
- **Handling Complex Garments:** Developing specialized algorithms or integrating advanced techniques to better handle garments with intricate drapes, accessories (like dupattas), and complex patterns will significantly improve the realism of the try-on experience. This might involve multi-stage deformation processes or AI models specifically trained on such complex garment types.
- **3D Virtual Try-On:** While the current project focuses on 2D, transitioning to 3D virtual try-on could provide a more immersive and accurate experience. This would involve creating 3D models of both the user and the garments, allowing for real-time manipulation and viewing from multiple angles.
- **Real-time AR Integration:** Integrating Augmented Reality (AR) capabilities would allow users to virtually try on clothes in real-time using their smartphone cameras, further enhancing convenience and engagement.
- **Automated Cloth Extraction Refinement:** Further research into fully automated and highly accurate cloth extraction from diverse commercial images could streamline the preprocessing pipeline and reduce reliance on manual selection.
- **User Feedback and Iteration:** Incorporating user feedback mechanisms to continuously refine the model and address perceived inaccuracies in fit or appearance.



Figure 3: Distortions due to extra cloth. In (a), the selected cloth contains the dupatta in hand, causing distortions. In (b), the person image has a dupatta in hand, leading to slight deformation of the hand and disruption in the shalwar.

## 7 Conclusion

The Virtual Fashion Fit project successfully developed an AI-driven 2D virtual try-on system specifically for traditional Pakistani clothing, utilizing and adapting the SD-VITON model. By focusing on crucial preprocessing steps like robust cloth extraction and optimized human parsing, the system delivers more realistic visualizations for traditional garments compared to generic solutions. While challenges such as fixed resultant cloth length and distortions with extra fabric elements remain, the project lays a strong foundation for enhancing the online retail experience for traditional fashion. Future work will focus on addressing these limitations and exploring advancements like 3D and AR integration to further bridge the gap between online visualization and physical try-on.

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